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Effects on rumen pH and feed intake of a dietary concentrate challenge in cows fed rations containing pH modulators with different neutralizing capacity

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ABSTRACT

Forty-five Holstein lactating cows (41 ± 8.8 kg/d of milk yield, 96 ± 35.6 days in milk, and 607 ± 80.4 kg of body weight) were enrolled in this study to assess the effects of diets supplemented with sodium bicarbonate or a magnesium-based product and their corresponding differences in dietary cation-anion difference (DCAD) on rumen pH, rumen microbial population, and milk performance of dairy cattle exposed to an induced decrease in rumen pH through a dietary challenge. Cows were randomly allocated to 3 total mixed rations (TMR) differing in the type of supplement to modulate rumen pH: (1) control, no supplementation; (2) SB, supplemented with 0.82% of sodium bicarbonate with a neutralizing capacity (NC) of 12 mEq/g; and (3) MG, supplemented with 0.25% of magnesium oxide (pHix-Up, Timab Magnesium) with a NC of 39 mEq/g. Thus, SB and MG rations had, in theory, the same NC. The 3 TMR differed for control, SB, and MG in their DCAD-S (calculated considering Na, K, Cl, and S), which was on average 13.2, 21.2, and 13.7 mEq/100 g, respectively, or DCAD-Mg (calculated accounting for Mg, Ca, and P), which was 31.4, 41.2, and 35.2 mEq/100 g, respectively. The study lasted 63 d, with the first 7 d serving as a baseline, followed by a fortnightly progressive decrease of dietary forage-to-concentrate ratio (FCR) starting at 48:52, then 44:56, then 40:60, and finishing at 36:64. Individual dry matter intake (DMI) was recorded daily. Seven cows per treatment were equipped with electronic rumen boluses to monitor rumen pH. Control and SB cows consumed less dry matter (DM; 23.5 ± 0.31 kg/d) than MG cows (25.1 ± 0.31 kg/d) when fed dietary FCR of 44:56 and 40:60. Energy-corrected milk decreased from 40.8 ± 1.21 to 39.5 ± 1.21 kg/d as dietary FCR decreased, independently of dietary treatments. Rumen pH decreased and the proportion of the day with rumen pH <5.8 increased as dietary FCR decreased, and at low dietary FCR (i.e., 36:64) rumen pH was greater in MG cows than in control and SB cows. Reducing the DCAD-S from 28 to 18 mEq/100 g or the DCAD-Mg from 45 to 39 mEq/kg had no effects on DMI or milk yield. Cows supplemented with ~62 g/d of magnesium oxide (pHix-Up) maintained a greater rumen pH and consumed more DM than cows supplemented with ~200 g/d of sodium bicarbonate when fed a diet with low FCR.

Key words: dietary cation-anion difference, intake, rumen acidosis

INTRODUCTION

The progressive genetic improvement of dairy cattle has led to continuous increases in milk production capacity. This high milk production potential can only be achieved if the diet that cows are able to consume supplies sufficient energy and nutrients. The high demand for energy is often met by rations with incremental amounts of NFC and, to some extent, by adding fat to the diet. However, increasing the amount of NFC is usually coupled with a reduction of forage provision, and both the high amounts of NFC, which may ferment vigorously in the rumen, and the low supply of forage may lead to an accumulation of fermentation end products, which may in turn decrease rumen pH (Blanch et al., 2010). The reduction of rumen pH may alter rumen microbiota (Ishaq et al., 2017), rumen fermentation (Stefanska et al., 2018), and overall animal performance (Danscher et al., 2015).

Several strategies can be used to control rumen pH, such as supplementing rations with neutralizing agents (Bach et al., 2018; Humer et al., 2018), live yeast (Thrune et al., 2009; Bach et al., 2019), or yeast

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cultures (Dias et al., 2018), among others. Regarding the use of neutralizing agents, several studies report increases in DMI (Erdman et al., 1982; Rogers et al., 1985; Staples et al., 1988) and milking performance (Kilmer et al., 1981; Thomas et al., 1984; Rogers et al., 1985) when supplementing rations with sodium bicarbonate. However, less data are available about the effects of magnesium oxide on the performance of dairy cows (Beede, 2017; Bach et al., 2018; Tebbe et al., 2018).

Dietary cation-anion difference (DCAD) is highly affected by sodium bicarbonate supplementation, and several authors have reported improvements in milk yield and DMI when DCAD increases (Sanchez and Beede, 1996; Hu and Murphy, 2004; Iwaniuk and Erdman, 2015). However, when relying on sources of magnesium oxide rather than sodium bicarbonate to control rumen pH, DCAD decreases and there are concerns that this could affect DMI and milk yield. Dietary cation-anion difference is commonly calculated considering dietary content of K, Na, Cl, and S (Jackson et al., 2001), but alternatively, DCAD can also be calculated by considering the dietary content of K, Na, Ca, Mg, P, Cl, and S (Goff et al., 1997). Herein, we hypothesized that removing sodium bicarbonate and supplying magnesium oxide would not have detrimental effects on DMI or milk yield. Furthermore, DCAD between diets containing sodium bicarbonate or magnesium oxide would not differ much when accounting for their content in Mg. Thus, the objective of this study was to evaluate the effects of diets supplemented with sodium bicarbonate or a magnesium-based product and their corresponding differences in DCAD on rumen pH, rumen microbial population, and milk performance of dairy cattle exposed to an induced decrease in rumen pH through a dietary challenge.

MATERIALS AND METHODS

All procedures involving animals were conducted under the supervision and approval of the Animal Care Committee of the Institute of Agrifood Research and Technology (Barcelona, Spain; expedient number: 11456).

Forty-five lactating (multiparous) cows were randomly split into 3 treatments (n = 15). Randomization was conducted using a random generator script in Python written by author AB (2020). Sample size was defined for milk fat yield as main outcome variable (as it was expected to collect the potential effects of DMI, milk yield, and rumen changes) based on a power analysis for repeated measures conducted with GLIMMPSE (www.glimmpse.samplesizeshop.org) and a Hotelling-Lawley trace test (Akbari et al., 2013) with power = 0.80, α = 0.05, and an expected standard deviation of 0.3 units. Cows were milked twice daily, and feed was provided twice daily with a targeted daily feed refusal rate of 3%. Farm personnel was blinded to treatments. Cows were housed in a single pen equipped with freestalls bedded every 2 d with chopped straw and equipped with 60 (20 per type of dietary supplement) electronic feed bins (MooFeeder, MooSystems) that controlled the access of cows in the same pen to specific bins containing the different dietary treatments. The electronic bins, in addition to controlling access to feed, recorded the time of day and the amount of feed consumed at every visit throughout the study.

The study followed a complete randomized design with 3 treatments. Cows were randomly allocated to 3 TMR differing in the type of supplement to modulate rumen pH: (1) a TMR with no supplementation (**control**), (2) a TMR supplemented with 0.82% of sodium bicarbonate (**SB**), and (3) a TMR supplemented with 0.25% of a blend of magnesium oxide (**MG**; pHix-Up, Timab Magnesium). Sodium bicarbonate was determined to have a neutralizing capacity (**NC**) of 12 mEq/g, and the blend of magnesium oxide was determined to have an NC of 39 mEq/g; thus both SB and MG were formulated to have an equivalent NC. The NC was assessed following Bach et al. (2018). Briefly, the milliequivalents of protons (from hydrochloric acid) required to lower the pH of a solution containing 2.5 g of magnesium or sodium bicarbonate in a volume of 50 mL to a pH of 3 starting from a pH of either 5.5 or 6.5 was determined.

During the study (conducted between June and September 2021), the amount of barley included in the ration was progressively increased. Starting after a 7-d basal period, on subsequent 14-d periods, 1 kg of forage was replaced by 1 kg of barley. The different TMR consumed by cows are depicted in Table 1. The control ration for period 1 was the same ration that was fed for all cows in all treatments during the 7-d basal or pretrial period. All grains were included in the TMR ground between 2 and 3 mm, and forages were chopped at a theoretical length of cut of 30 mm. Samples of TMR were taken at weekly intervals. Feed samples were stored at −20°C and composited for subsequent nutrient analysis. All rations were formulated to be isoenergetic and isonitrogenous within dietary forage-to-concentrate ratio (**FCR**), but there were some differences in ingredient composition, with the SB diets having a slightly lower amount of wheat and barley than the rest due to the inclusion of sodium bicarbonate and still reaching the energy and protein targets.

Individual milk production at every milking was determined using electronic milk meters (AfiMilk, Afikim Ltd.), and milk fat and milk protein content

Table 1. Nutrient and chemical compositions of the total mixed ration for each dietary treatment and period[1]

	Period 1			Period 2			Period 3			Period 4		
Item	Control	SB	MG	Control	SB	MG	Control	SB	MG	Control	SB	MG
Ingredient, % of DM												
Alfalfa hay	19.9	19.9	19.9	19.9	19.9	19.9	19.9	19.9	19.9	19.9	20.0	19.9
Fescue hay	14.5	14.5	14.5	10.9	10.8	10.9	7.23	7.23	7.24	3.62	3.64	3.62
Oat hay	7.23	7.95	7.24	7.25	7.95	7.24	7.23	7.95	7.24	7.23	7.29	7.24
Straw	5.54	5.55	5.55	5.56	5.54	5.55	5.54	5.55	5.55	5.55	5.59	5.55
Barley	4.12	3.83	4.05	7.50	7.48	7.71	11.4	11.1	11.3	15.0	14.8	14.9
Corn	33.9	33.9	33.7	34.0	33.9	33.7	33.9	33.9	33.7	33.9	34.2	33.7
Soybean meal	12.7	11.9	12.9	12.8	11.9	12.9	12.7	11.9	12.9	12.7	12.0	12.9
Wheat	0.72	0.29	0.58	0.72	0.29	0.58	0.72	0.29	0.58	0.72	0.29	0.58
Palm oil	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08	0.08
Sodium chloride	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33	0.33
Calcium carbonate	0.82	0.74	0.78	0.74	0.74	0.78	0.82	0.74	0.78	0.82	0.75	0.78
pHix-up[2]	—	—	0.25	—	—	0.25	—	—	0.25	—	—	0.25
Sodium bicarbonate	—	0.82	—	—	0.82	—	—	0.82	—	—	0.82	—
Premix[3]	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21	0.21
Nutrient, DM basis												
CP, %	15.9	15.5	15.9	16.0	15.6	16.0	16.0	15.7	16.1	16.1	15.8	16.2
NE_L Mcal/kg	1.66	1.65	1.66	1.67	1.66	1.67	1.69	1.68	1.69	1.71	1.70	1.70
NDF, %	32.2	32.2	32.2	30.8	30.9	30.7	29.3	29.5	29.3	27.9	27.9	27.9
ADF, %	19.0	19.1	19.0	18.0	18.1	17.9	16.9	17.0	16.9	15.8	15.8	15.8
Fat, %	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6	2.6
Ash, %	7.3	8.1	7.6	7.0	7.80	7.3	6.8	7.5	7.0	6.6	7.3	6.8
NFC, %	42.0	41.5	41.8	43.6	43.1	43.4	45.3	44.7	45.0	46.9	46.5	46.6
Ca, %	0.85	0.84	0.85	0.84	0.81	0.83	0.82	0.79	0.81	0.80	0.78	0.79
P, %	0.34	0.34	0.36	0.33	0.32	0.34	0.36	0.34	0.33	0.36	0.33	0.36
Mg, %	0.21	0.21	0.35	0.21	0.21	0.35	0.21	0.21	0.34	0.20	0.21	0.34
Na, %	0.17	0.39	0.17	0.17	0.39	0.17	0.16	0.39	0.17	0.16	0.40	0.16
K, %	1.61	1.62	1.61	1.53	1.52	1.53	1.45	1.44	1.47	1.37	1.35	1.38
S, %	0.24	0.22	0.24	0.23	0.23	0.23	0.23	0.22	0.22	0.22	0.24	0.22
Cl, %	0.53	0.50	0.53	0.52	0.52	0.51	0.50	0.51	0.50	0.49	0.50	0.49
DCAD-S, mEq/100 g	18.6	30.6	18.6	17.5	26.8	17.8	15.6	25.7	17.2	14.4	22.8	14.7
DCAD-Mg, mEq/100 g	34.1	45.8	37.2	32.9	41.8	36.3	30.1	40.0	35.3	28.3	37.1	31.9
Forage, %	47.1	47.9	47.2	45.1	44.2	43.5	39.8	40.6	39.9	36.2	36.6	36.3

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. The 3 TMR differed for control, SB, and MG in their DCAD-S (calculated considering Na, K, Cl, and S) or DCAD-Mg (calculated accounting for the addition of Mg, Ca, and P).

[2]Timab Magnesium.

[3]Premix contained 81.6 mg/kg of Zn; 11.5 mg/kg of Cu; 57.6 mg/kg of Mn; 9.86 mg/kg of Co; 1.92 mg/kg of I; 0.34 mg/kg of Se; 58 mg/kg of S; 120,000 IU/kg of vitamin A; 28,800 IU/kg of vitamin D; and 1,920 IU/kg of vitamin E.

were also determined electronically every milking using the AfiLab system (Afikim Ltd.), which was calibrated fortnightly. All cows were weighed on a daily basis at the exit of the milking parlor using an electronic scale.

Seven cows per treatment were randomly selected before the start of the study and received a rumen pH meter bolus (SmaXtec Animal Care Sales GmbH) programmed to record rumen pH at 10-min intervals. The number of cows to be sampled was based on a power analysis for repeated measures as described above but using a power = 0.80, $\alpha = 0.05$, and an expected standard deviation of 0.21 units. At d 0 and every 10 d, a sample of rumen fluid was collected from 7 cows per treatment (also randomly selected before the onset of the study) at 2 h after the morning feeding using a stomach tube. Samples were frozen at −80°C until subsequent DNA extraction and processing. The DNA from rumen fluid was extracted and purified using the DNeasy PowerSoil Pro isolation kit (Qiagen) following the procedure suggested by the manufacturer. The DNA concentrations and their purity were measured by spectrophotometry using a NanoDrop ND-1000 UV/Vis Spectrophotometer (NanoDrop Technologies Inc.) and the Qubit Fluorometer (Life Technologies). One microgram of DNA was analyzed following the 1D native barcoding genomic DNA SQK-LSK109 (with EXP-NBD104 and EXP-NBD114) and ligation sequencing kit (SQK-LSK109) protocol from Oxford Nanopore Technologies using the GridION sequencer (Oxford Nanopore Technologies) and R9.4.1 Flow Cells (Oxford Nanopore Technologies). Guppy toolkit (Version 6.1, Oxford Nanopore Technologies) was used for basecalling and demultiplexing. Average and standard deviation of sequencing depth was 407,161 ± 82,031 reads per sample. A quality control was then applied removing sequences with quality score <7 and length

<150 bp. Sequence analysis was performed using the SqueezeMeta pipeline (sqm_longreads, version 1.6.0) for long reads (Tamames and Puente-Sánchez, 2019), which performs Diamond Blastx searches against the National Center for Biotechnology Information's GenBank (downloaded May 12, 2021, from www.ncbi.nlm.nih.gov) taxonomic database, then identifies and annotates open reading frames using the lowest common ancestor method. This tool is specifically designed to process long reads from Oxford Nanopore Technologies. Reads were processed in Blastx by SqueezeMeta longreads pipeline. All sequences mapped as nonmicrobial (i.e., viruses, animals, and plants) were discarded. Microbial sequences were then filtered by prevalence to reduce data sparsity and sequencing errors.

Every 10 d, grab fecal samples were obtained directly from the rectums of 7 cows per treatment at 2, 4, and 8 h relative to morning feeding. Samples were composited within cow and day to subsequently measure acid-insoluble ash and NDF contents and later calculate apparent total-tract NDF digestibility (Van Keulen and Young, 1977). Again, the number of cows to be sampled was based on a power analysis for repeated measures with power = 0.80, α = 0.05, and expected standard deviation for NDF apparent digestibility of 1.5 units. Fecal pH was measured at each occasion immediately after collection. All cows were blood-sampled from the jugular vein using vacutainers without additives (Beckton Dickinson) at d 0 and every 10 d throughout the study. Samples were kept refrigerated until being centrifuged at 1.50 × g for 10 min at room temperature and stored in aliquots at −20°C until subsequent serum analysis of Ca, Cl, Na, K, P, Mg, and haptoglobin. Serum concentrations of Ca, P, and Mg were determined using the arsenazo III method (Michaylova and Ilkova, 1971) and the phosphomolybdate method, and by direct colorimetry using the xylidyl blue reaction, respectively; those of Cl, Na, and K were determined using ion-selective electrodes (Levy, 1981); and serum haptoglobin concentrations were determined using a colorimetric method (Tridelta Development Ltd.).

Finally, all cows were urine-sampled by vulval stimulation at d 0 and every 10 d at 2, 4, and 6 h after the morning feeding and 2 h after the afternoon feeding. Urine pH was immediately (within 5 min of collection) measured using a glass-electrode pH meter, which was calibrated daily at room temperature using buffer solutions of pH 4.01 and 7.00. The glass electrode was rinsed using deionized water and dried prior to each use. At the end of the day, the 4 urine samples from each cow were composited and frozen until subsequent analysis of creatinine, Ca, and Mg. Urine creatine was determined by kinetic colorimetric analyses based upon the Jaffe reaction (Seaton and Ali, 1984). Urine Ca and Mg concentrations were determined using the arsenazo III method (Michaylova and Ilkova, 1971) and by direct colorimetry using xylidyl blue, respectively.

Energy-corrected milk was calculated following NRC (1989), as

$$\text{ECM (kg/d)} = 12.55 \times \text{fat yield (kg/d)} + 7.39 \times \text{protein yield (kg/d)} + 5.34 \times \text{lactose yield (kg/d)}.$$

Dietary cation-anion difference was calculated following Jackson et al. (2001), as

$$\text{DCAD-S (mEq/100 g)} = \{[\text{Na (g/kg)}/0.0023] + [\text{K (g/kg)}/0.00391]\} - \{[\text{Cl (g/kg)}/0.00355] + [\text{S (g/kg)}/0.00321] \times 2\}.$$

Also, DCAD-Mg (mEq/100g) was calculated following Goff et al. (1997), as

$$\text{DCAD-Mg (mEq/100 g)} = \{[\text{Na (g/kg)}/0.0023] + [\text{K (g/kg)}/0.00391] + [0.38 \times \text{Ca (g/kg)}/0.0020] + [\ 0.30 \times \text{Mg (g/kg)}/0.0012]\} - \{[\text{Cl (g/kg)}/0.00355] + [\text{S (g/kg)}/0.00321] \times 2 + [0.30\ 0.50 \times \text{P (g/kg)}/0.0017]\}.$$

Alpha diversity (i.e., diversity within a community from an animal sample) at the genus level was calculated as the inverse Simpson index with the phyloseq (McMurdie and Holmes, 2013) function estimate_richness() on R (Version 4.1.3; R Core Team, 2022). Beta diversity (i.e., diversity between communities of different animal samples) across all samples was calculated using the Bray-Curtis dissimilarity index (Bray and Curtis, 1957) considering the Euclidian distance in microbial composition between samples, and a gradient analysis for each treatment was performed using nonmetric multidimensional scaling also at the genus level. Thus, given the distance metric used herein, a high β-diversity indicates a low degree of similarity in specific taxa between individuals within a given dietary treatment or dietary FCR.

Because treatments were applied at the animal level, the experimental unit was the animal. Data from the 7-d baseline period was averaged within cow and used as a covariate, and the first 7 d of each period were excluded from the analysis. All data were checked for normality and transformed to a normal distribution when original data did not follow a normal distribution (according to the Shapiro-Wilk test), as was the case for serum haptoglobin concentrations. Data were submitted to a

Table 2. Performance as affected by a neutralizing agent and dietary forage-to-concentrate ratio

	Neutralizing agent[1]				P-value[2]		
Item	Control	SB	MG	SE	NA	FC	NA×FC
DMI, kg/d	24.1	24.0	24.8	0.40	0.28	0.06	0.05
Eating time, min/d	193[b]	221[a]	215[a]	4.40	<0.01	<0.01	<0.01
Eating rate, g of DM/min	130[a]	112[c]	118[b]	2.56	<0.01	<0.01	<0.01
Milk yield, kg/d	38.5	38.9	38.5	0.58	0.84	<0.01	0.07
Milk fat, %	3.59	3.67	3.61	0.05	0.45	0.03	0.07
Milk fat, kg/d	1.37	1.43	1.38	0.02	0.09	<0.01	0.29
Milk protein, %	3.38	3.35	3.35	0.01	0.08	<0.01	0.44
Milk protein, kg/d	1.30	1.30	1.29	0.02	0.93	<0.01	0.43
ECM, kg/d	39.8	40.4	40.5	2.05	0.96	<0.01	0.09
Feed efficiency[2]	1.67	1.70	1.66	0.09	0.95	0.19	0.04

[a–c]Values within row with uncommon superscripts differ at $P < 0.05$.

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide.

[2]NA = effect of neutralizing agent; FC = effect of dietary forage-to-concentrate ratio; NA×FC = effect of the interaction between dietary neutralizing agent and forage-to-concentrate ratio.

[3]Calculated as ECM/DMI.

mixed-effects model that included the fixed effects of treatment, period, and their 2-way interactions, plus the random effect of cow using SAS (Version 9.4, 2016, SAS Institute Inc.). Repeated structure was modeled using animal within treatment as a subject and an autoregressive order 1 covariate-variance matrix structure. Least squares means (± SEM) are reported. Mean separation was conducted using the Tukey method.

For rumen pH records, data from the last 7 d of each period were averaged within cow and hour of the day, and a model that accounted for the fixed effects of treatment, period, hour, and their 2- and 3-way interactions—including hour of the day as a repeated measure, total daily DMI as a covariate, and cow within period and treatment as a random effect—was run using an autoregressive order 1 covariate-variance matrix, which yielded the smallest Bayesian information criterion among the tested covariate-variance structures. The interactions between period and hour and among treatment, period, and hour were not significant ($P > 0.18$); thus, all data were averaged by cow, treatment, and hour, and a mixed-effects model that accounted for the fixed effects of treatment, hour, and their 2-way interaction was run, including cow within treatment as a random effect.

Beta diversity differences among treatments were assessed by permanova (a nonparametric multivariate statistical analysis using permutations to avoid bias) on Bray-Curtis dissimilarity matrices performed with 1,000 permutations. Differential relative abundance among treatments was assessed using a multivariate ANOVA with the Limma package (Ritchie et al., 2015) of R (Version 4.1.3; R Core Team, 2022) using centered, log-transformed data as input and correcting for false discovery rates (Benjamini and Hochberg, 1995). False discovery rate-adjusted P-values are shown throughout the manuscript.

RESULTS AND DISCUSSION

Feed Intake and Feeding Behavior

Dry matter intake tended ($P = 0.06$) to decrease from 24.9 ± 0.31 to 24.0 ± 0.31 kg/d as the dietary FCR decreased (Table 2). Control and SB cows consumed less ($P = 0.046$) DM (23.5 ± 0.31 kg/d) than MG cows (25.1 ± 0.31 kg/d) when dietary FCR was 44:56 and 40:60 (Figure 1). The time cows devoted to eat was affected by an interaction between the dietary neutralizing agent and dietary FCR, with cows on control showing a lower eating time than those on SB or MG during periods 2, 3, and 4 (Figure 2). Furthermore, the daily amount of time that cows devoted to eating decreased from 230 ± 3.64 to 192 ± 3.69 min/d as dietary FCR decreased, likely due to the concomitant decrease in DMI. As a consequence, the eating rate of control cows was greater than that of cows on SB and MG as dietary FCR decreased beyond 48:52 (Figure 3).

Dietary FCR has been previously reported to affect DMI. Most studies report an increase in DMI when the dietary FCR decreases, which is opposite of the findings herein. For example, Kargar et al. (2010) reported a 2-kg increase in DMI when FCR decreased from 34:66 to 45:55, and both dietary NDF and the net energy of lactation concentrations were maintained constant between the 2 diets. However, Olijhoek et al. (2018) also reported an increase of about 2 kg of DMI when reducing FCR from 68:32 to 39:61, but in this case, dietary NDF, starch, and energy content differed between diets. Conversely, Aguerre et al. (2011) reported no changes

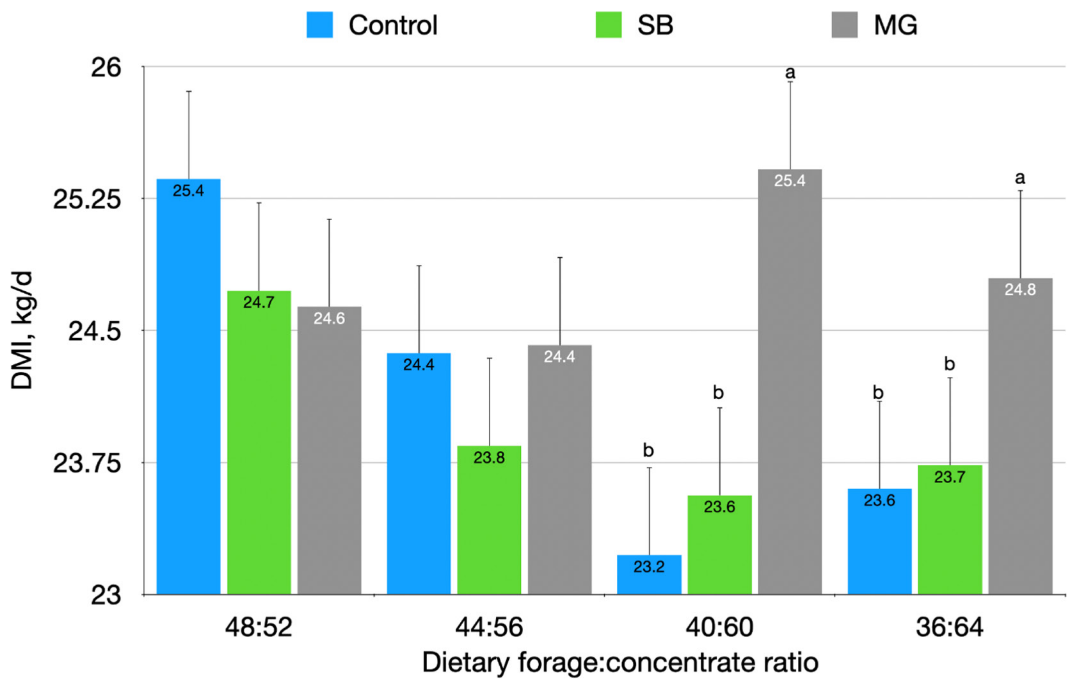

Figure 1. Dry matter intake as affected by neutralizing agent and dietary forage-to-concentrate ratio. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

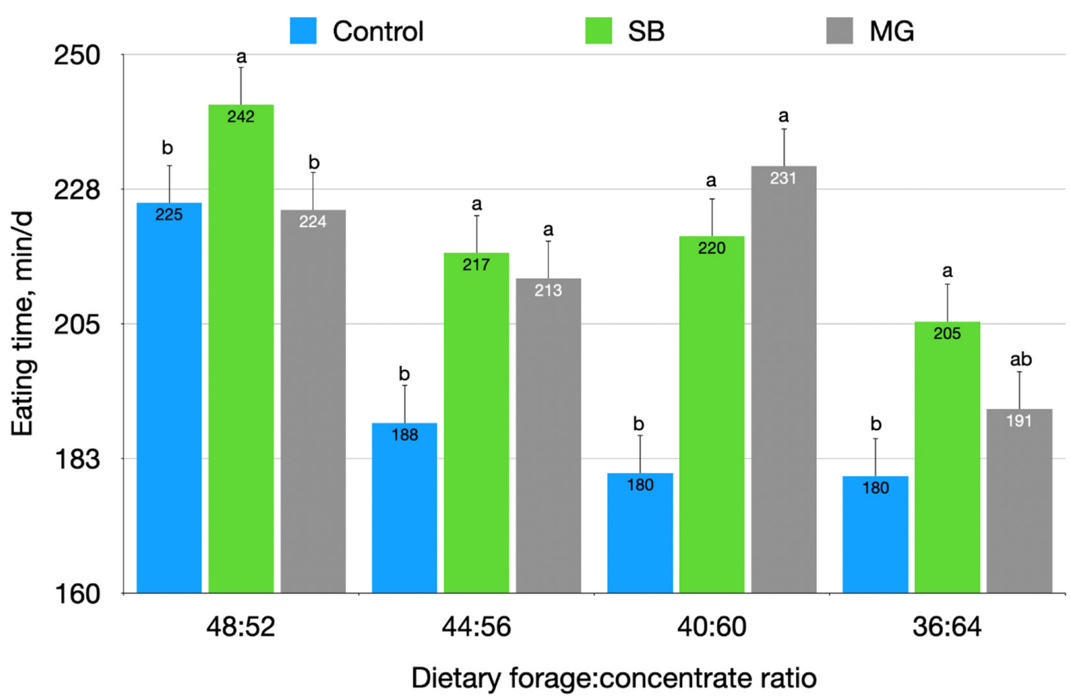

Figure 2. Time devoted to eat as affected by neutralizing agent and dietary forage-to-concentrate ratio. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

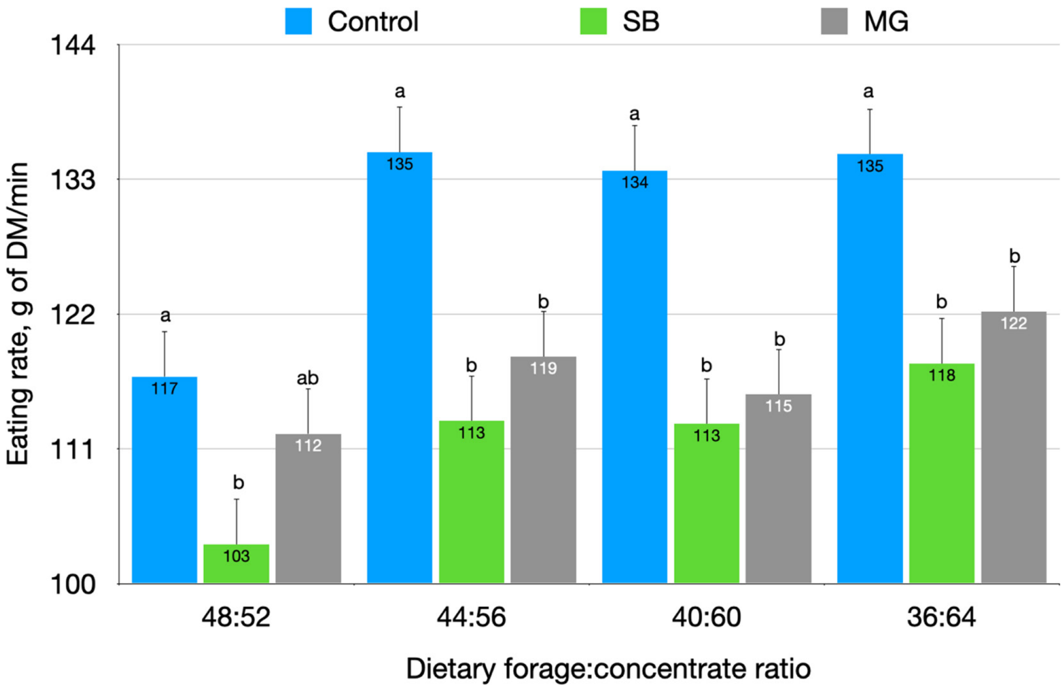

Figure 3. Eating rate as affected by neutralizing agent and dietary forage-to-concentrate ratio. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

in DMI when comparing rations containing FCR of 68:32 and 47:53. In the current study, differences in dietary FCR were accomplished by sudden progressive replacements of forage by barley in an attempt to cause a mild disruption in the rumen environment. Starch from barley is rapidly fermented in the rumen (Offner et al., 2003), and Oba and Allen (2003) showed that provision of rapidly fermentable starch reduces DMI by promoting satiety and reducing meal sizes and increasing eating rate. Furthermore, in the current study, the reduction of dietary FCR resulted in a decrease in rumen pH, and rumen acidosis has been shown to hamper DMI (Gao and Oba, 2016). In this study, the eating rate was increased as dietary FCR decreased, which would further support the role of rapidly fermentable carbohydrates on modulation of DMI. Greter et al. (2008) also reported an increased eating rate as dietary FCR decreased in dairy heifers, and Friggens et al. (1998) and DeVries et al. (2009) also found greater eating rates when FCR decreased in dairy cattle. Interestingly, feeding SB or MG voided the effect of dietary FCR on increasing the eating rate when the dietary FCR was lowest (36:64), which may suggest that eating rate is not only controlled by dietary FCR or diet fermentability, but it may also be influenced by ruminal conditions. In fact, DeVries et al. (2009) described a decrease in eating rate when cows experienced rumen acidosis. Herein, MG cows showed a marked increase in eating rate when dietary FCR was 36:64, whereas cows on control or SB did not, which could be a consequence of a more stable rumen pH observed in MG than in SB and control cows as discussed below.

Some in the field hesitate to remove sodium bicarbonate from the diets because it causes a reduction in DCAD (as less Na is supplied by the ration) and some studies have reported a positive (quadratic) relationship between DCAD-S and DMI (Hu and Murphy, 2004; Hu et al., 2007; Iwaniuk and Erdman, 2015). In the current study, DCAD-S of SB was 10 mEq/100 g or about 60% greater than in control and MG treatments. According to the equation from Iwaniuk and Erdman (2015), such a difference in DCAD-S should result in about 300 g/d additional DMI in SB than in control or MG treatments. However, overall DMI did not differ among treatments, and it was actually greater in MG cows when dietary FCR was 40:60 or 36:54 (Figure 1). These results would suggest that a potential decrease in DMI caused by a lower DCAD-S in MG than in SB was overcome by a positive effect of an improved rumen environment on feed intake. Furthermore, DCAD-Mg was only 14% lower in MG and 24% lower in control than in SB, and thus it may just be that such a small difference (14%) has no consequences on feed intake, in which case, the use of DCAD-Mg may be preferred over DCAD-S when feeding rations with magnesium oxide and without sodium

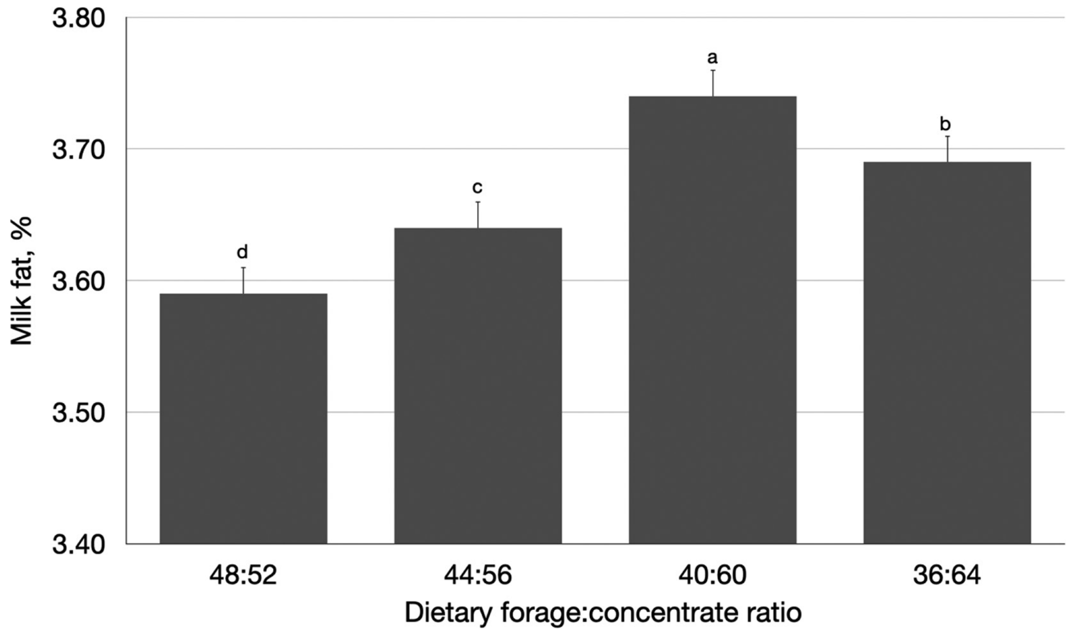

Figure 4. Milk fat content as affected by dietary forage-to-concentrate ratio. [a-d]Bars with uncommon letters differ at $P < 0.05$. Error bars denote SEM.

bicarbonate and attempting to infer potential effects on cation-anion balance and DMI.

Milking Performance

Milk yield decreased from 39.8 ± 0.40 to 37.6 ± 0.40 kg/d and ECM decreased from 40.8 ± 1.21 to 39.5 ± 1.21 kg/d as dietary FCR decreased independently of dietary treatments (Table 2). Milk fat content increased from 3.59 ± 0.03% in period 1 to 3.74 ± 0.03% in period 3, and then decreased to 3.69 ± 0.03% in period 4, independently of dietary treatments (Figure 4). Despite the increase in milk fat content, milk fat yield decreased from 1.42 ± 0.02 to 1.38 ± 0.02 kg/d as dietary FCR decreased (Table 2). Milk fat content and yield are typically reduced when the dietary FCR decreases (Macleod et al., 1983; Argov-Argaman et al., 2014; Machado et al., 2014). However, milk protein content increased from 3.27 to 3.46% (Figure 5) and milk protein yield from 1.30 ± 0.01 to 1.33 ± 0.01 kg/d (data not shown) as dietary FCR decreased. Several studies have reported that increases in dietary NFC concentrations result in increased milk protein content and yield (Batajoo and Shaver, 1994; Oliveira et al., 2020).

Feed efficiency (**FE**) was unaffected by treatments, but there was an interaction between treatment and FCR, as FE progressively decreased in MG cows as FCR decreased (Figure 6). This change in FE reflected the increase in DMI in MG as FCR decreased (Figure 1). Perhaps a longer exposure to treatments may have allowed the cows to respond to differences in DMI in terms of milk yield or body accretion.

Rumen pH

Average rumen pH was influenced by an interaction ($P < 0.01$) between dietary FCR and a neutralizing agent that resulted in lower rumen pH values in SB than in control and MG cows when dietary FCR was 44:56 and lower values in control and SB than in MG cows when dietary FCR was 36:64 (Figure 7). Also, average rumen pH decreased ($P < 0.01$) from 6.25 ± 0.04 to 5.85 ± 0.04 as dietary FCR decreased (Table 3). Similarly, the proportion of time that rumen pH of cows was below 5.8 (a threshold often considered as a situation of subacute rumen acidosis) was greater ($P < 0.05$) in SB than in control and MG cows when dietary FCR was 36:64 (Figure 8). Furthermore, the proportion of time that rumen pH was below 5.8 increased ($P < 0.05$) from 23.2 ± 5.3% to 31.6 ± 5.3% as dietary FCR decreased (Table 3). Despite the fact that, in theory, the NC of the SB diet should be equivalent to that of the MG treatment (because the amounts of sodium bicarbonate and magnesium oxide were adjusted based on their respective NC), MG was able to maintain a greater average rumen pH throughout the day and diminish the amount of time with a rumen pH <5.8, suggesting a more sustained effect of MG in the rumen compared with SB. In fact, when pooling all pH

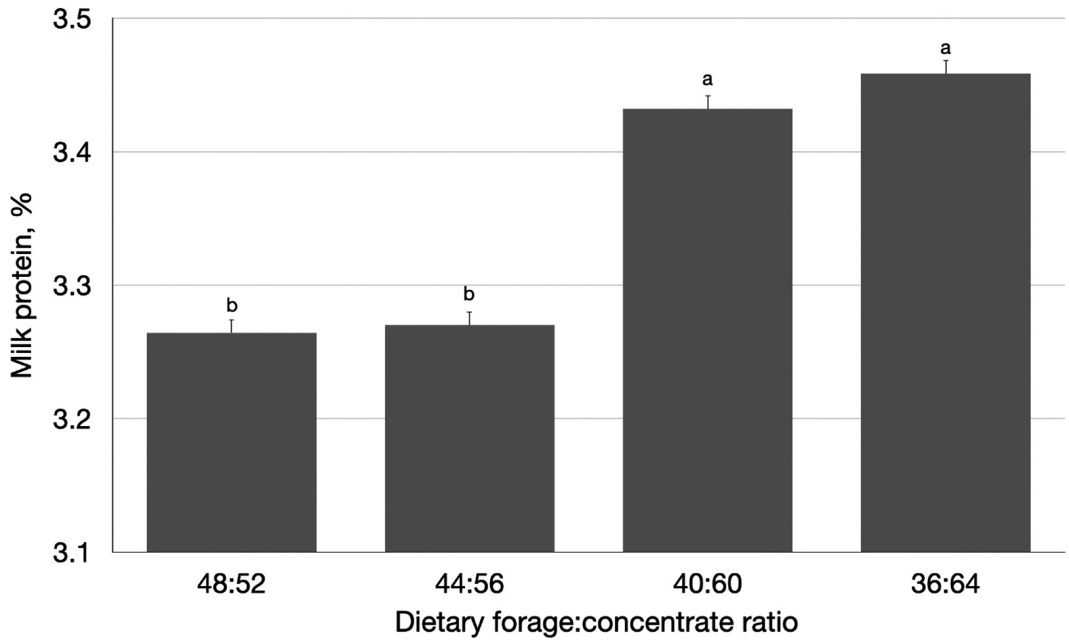

Figure 5. Milk protein content as affected by neutralizing agent and dietary forage-to-concentrate ratio. [a,b]Bars with uncommon letters differ at $P < 0.05$. Error bars denote SEM.

values for the last 7 d from each period and averaging them by hour of the day, it can be seen that SB cows had the greatest rumen pH during the first 2 to 3 h after the morning feeding (Figure 9), but then, starting about 4 h after the morning feeding, MG cows had the greatest rumen pH for the remainder of the day, with control cows showing intermediate values. A potential reason for control cows to have slightly greater rumen pH values than SB cows could be that control cows consumed less feed, and thus less fermentable matter, than SB cows. Figure 9 also depicts the cumulative DMI throughout the day and illustrates that MG cows consumed more feed than control and SB cows during the first hour after the morning feeding. Also, MG cows had a greater feed consumption at 1600 and 1700 h than control cows, with SB showing intermediate cumulative intakes.

Digestibility and Metabolism

There was an interaction ($P = 0.03$) between dietary FCR and the neutralizing agent on apparent total-tract NDF digestibility, with control and SB cows having lower digestibility than MG cows when fed a dietary FCR of 36:64 (Figure 10). Furthermore, apparent total-tract NDF digestibility decreased ($P < 0.01$) from 54.6 ± 1.37% to 50.1 ± 1.37%, to 46.6 ± 1.37%, to 28.3 ± 1.37% as dietary FCR changed from 48:52 to 44:56, to 40:60, to 36:64, respectively. Reductions in NDF digestibility as dietary NFC content

Table 3. Rumen, urine, and fecal pH as affected by dietary neutralizing agent and dietary forage-to-concentrate ratio

	Neutralizing agent[1]				P-value[2]		
Item	Control	SB	MG	SE	NA	FC	NA×FC
Rumen pH	6.01	5.99	6.11	0.07	0.46	<0.01	<0.01
Proportion of the day with rumen pH <5.8, %	26.8	32.3	23.3	0.09	0.76	0.02	0.04
Urine pH	7.96[b]	8.04[a]	7.95[b]	0.02	<0.01	<0.01	0.59
Fecal pH	6.31[b]	6.29[b]	6.50[a]	0.05	<0.01	<0.01	0.06

[a,b]Values within row with uncommon superscripts differ at $P < 0.05$.

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide.

[2]NA = effect of neutralizing agent; FC = effect of dietary forage-to-concentrate ratio; NA×FC = effect of the interaction between dietary neutralizing agent and forage-to-concentrate ratio.

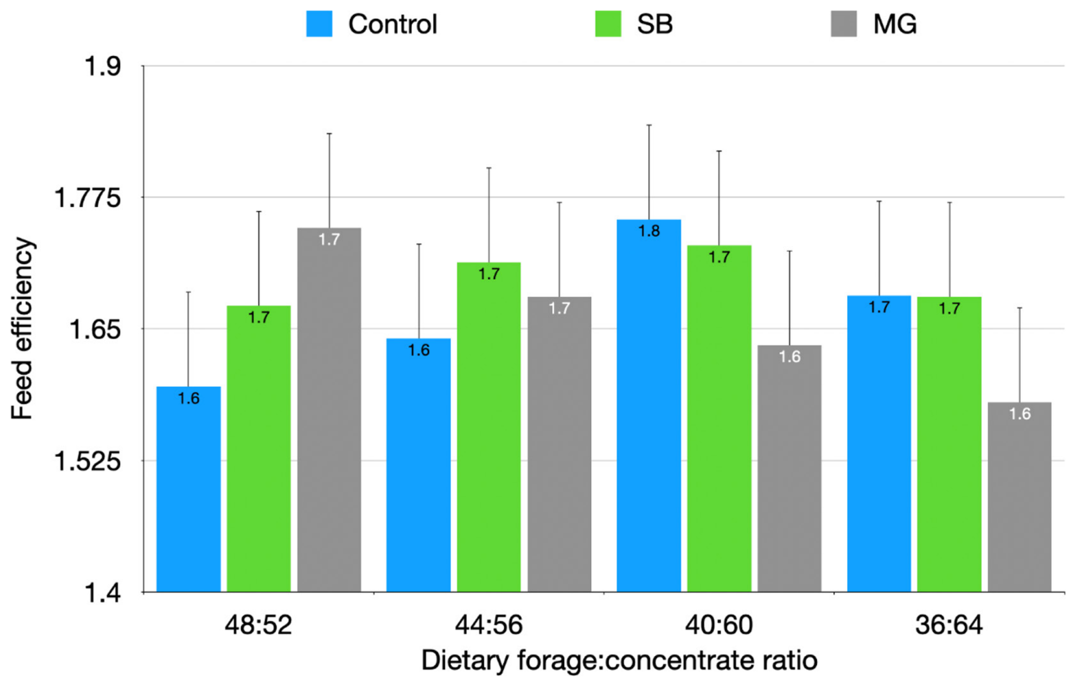

Figure 6. Feed efficiency as affected by neutralizing agent and dietary forage-to-concentrate ratio (interaction between neutralizing agent and dietary forage-to-concentrate ratio: $P < 0.05$). Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. Error bars denote SEM.

increases have been previously reported (Batajoo and Shaver, 1994). It is thought that this reduction is due to (1) a low rumen pH (Mould et al., 1983; Hoover, 1986; Calsamiglia et al., 2008) and (2) a potential increase in the passage rate resulting from a potential increase in DMI (Kargar et al., 2010; Olijhoek et al.,

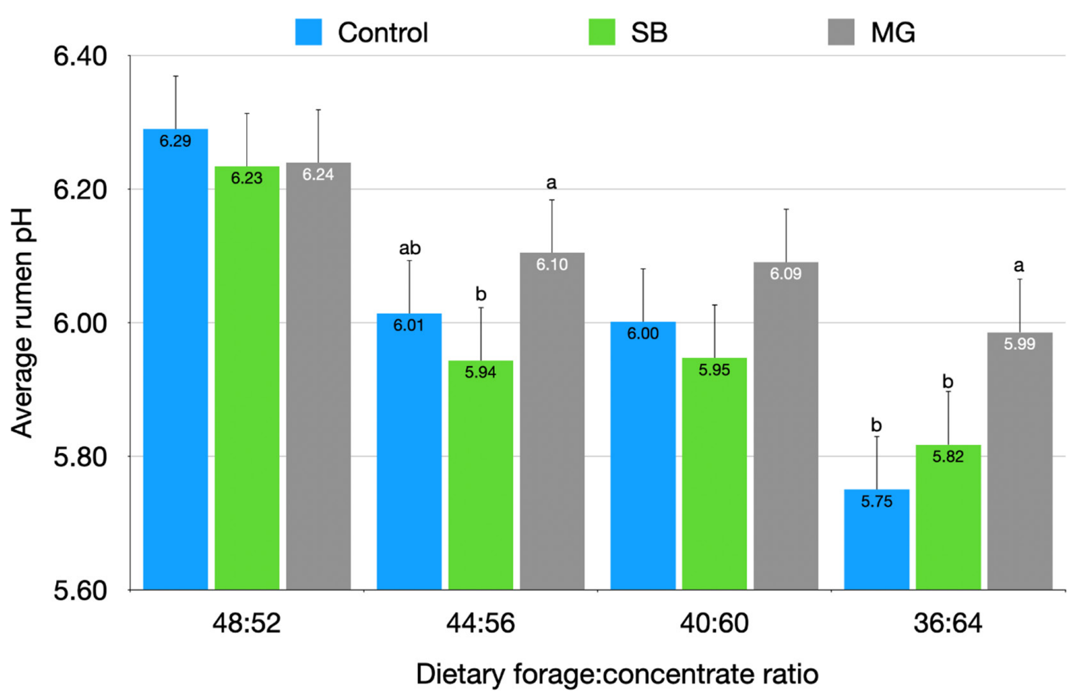

Figure 7. Average rumen pH as affected by neutralizing agent and dietary forage-to-concentrate ratio. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

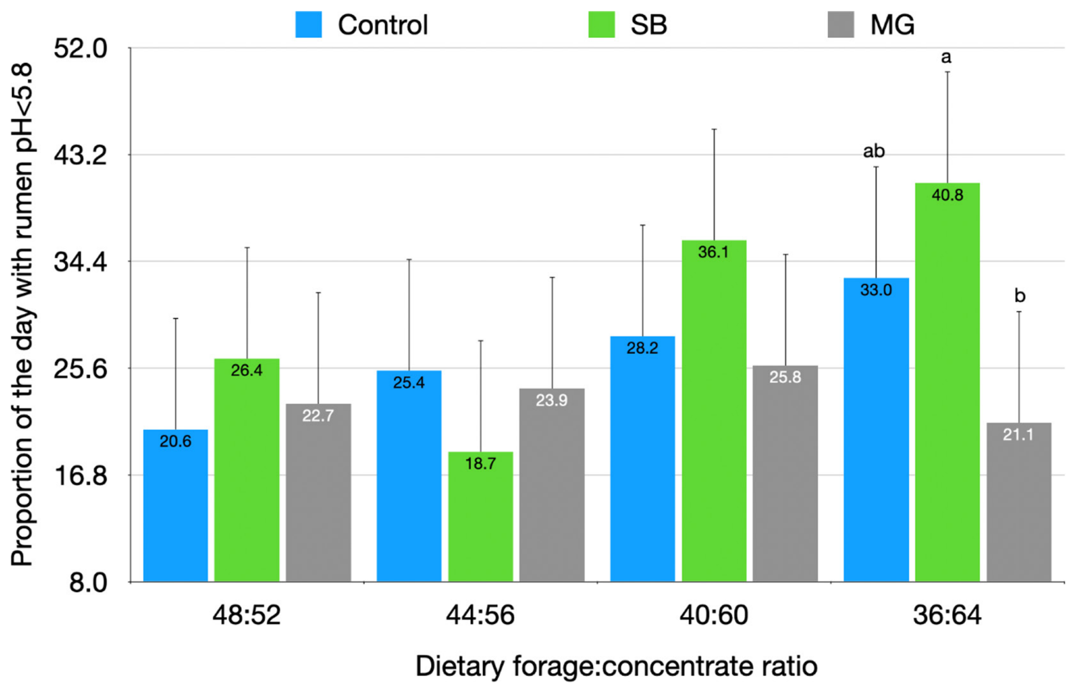

Figure 8. Effect of neutralizing agent and dietary forage-to-concentrate ratio on the proportion of time with rumen pH <5.8. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

2018). In the current study, DMI did not increase; thus, the reduction in apparent total-tract NDF digestibility could not be attributed to an increased passage rate. The reduction in fiber digestion at low rumen pH has been associated with a reduction in the fibrolytic microbial population mainly caused by an impaired ability of fibrolytic bacteria to attach to feed particles (Cheng et al., 1980) and to the slow replication rate of fibrolytic bacteria under low pH (Russell and Dombrowski, 1980). Herein, the greater apparent total-tract NDF digestibility in MG compared with control or SB cows with a dietary FCR of 36:64 (or a high NFC) could be partially attributed to a relatively greater rumen pH (Figures 7 and 8). This hypothesis is supported by Yang et al. (2001), who reported no change in apparent total-tract NDF digestibility when comparing 2 dietary FCR when rumen pH did not differ between dietary treatments.

Urine pH was greatest ($P < 0.01$) in SB cows independently of dietary FCR (Table 3), and decreased from 8.03 ± 0.02 to 7.96 ± 0.02 as dietary FCR decreased from 48:52 to 36:64. Studies reporting changes in rumen and urine pH are inconclusive. In some such studies, cows experiencing rumen acidosis had a decrease in urine pH (Danscher et al., 2015), and, in others, cows with rumen acidosis had increased urine pH (Li et al., 2012). It seems that, as concluded by Enemark et al. (2004), urine pH may not be suitable for predicting low ruminal pH due to the lack of a consistent relationship between the 2 parameters. Regarding the increased urine pH in SB cows, reports in the literature are also inconsistent. Stokes et al. (1986) described no changes in urine pH in cows fed rations with no supplementation, 0.7% sodium bicarbonate, or 0.7% sodium bicarbonate and 0.28% magnesium oxide. However, Martins et al. (2021) reported an increase in urine pH when supplementing dairy cows with both sodium bicarbonate (0.71% of the DM) and magnesium oxide (0.24% of the DM). Finally, Constable et al. (2019) described a negative relationship between urine Ca concentration and urine pH. In the current study, urine Ca concentrations were lowest in SB cows (Table 4), but urine pH was greatest in these cows (Table 3); thus, perhaps the increased urine pH may have been linked to changes in Ca metabolism induced by sodium bicarbonate rather than changes in rumen pH, as further discussed below.

Conversely, fecal pH was greatest in MG cows (Table 3), and it was greatest with the diet containing an FCR of 44:56. Previous studies have reported increases in fecal pH when supplementing magnesium oxide in the diet (Erdman et al., 1982; Thomas et al., 1984; Stokes et al., 1986).

Serum Ca concentration was consistently lower in SB and MG than in control cows (Table 4), and it de-

Table 4. Concentration of selected minerals and haptoglobin in serum as affected by neutralizing agent and dietary forage-to-concentrate ratio

Item	Neutralizing agent[1]			SE	P-value[2]		
	Control	SB	MG		NA	FC	NA×FC
Calcium, mM	2.39[a]	2.33[b]	3.32[b]	0.02	0.02	0.03	0.31
Chloride, mM	96.6[ab]	97.4[a]	96.3[b]	0.29	0.03	<0.01	0.85
Phosphorus, mM	1.93	1.91	1.91	0.05	0.92	0.07	0.23
Magnesium, mM	0.88	0.90	0.89	0.01	0.68	0.62	0.95
Potassium, mM	4.35	4.32	4.26	0.05	0.23	0.06	0.07
Sodium, mM	137.3[a]	137.7[a]	136.5[b]	0.25	<0.01	<0.01	0.70
Haptoglobin, mg/dL	0.18	0.24	0.24	0.05	0.29	0.82	0.95

[a,b]Values within row with uncommon superscripts differ at $P < 0.05$.

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide.

[2]NA = effect of neutralizing agent; FC = effect of dietary forage-to-concentrate ratio; NA×FC = effect of the interaction between dietary neutralizing agent and forage-to-concentrate ratio.

creased from 2.37 ± 0.02 to 2.33 ± 0.02 mM as dietary FCR decreased (Table 4) independently of treatments. Opposite to the observations herein, a couple studies (Kilmer et al., 1981) reported a tendency or a significant decrease in serum Ca concentrations when supplementing dairy cows with sodium bicarbonate. However, Erdman et al. (1980) and Rogers et al. (1985) also reported a decrease in serum Ca concentrations when supplementing sodium bicarbonate, as observed herein. The decrease in serum Ca concentrations as the dietary

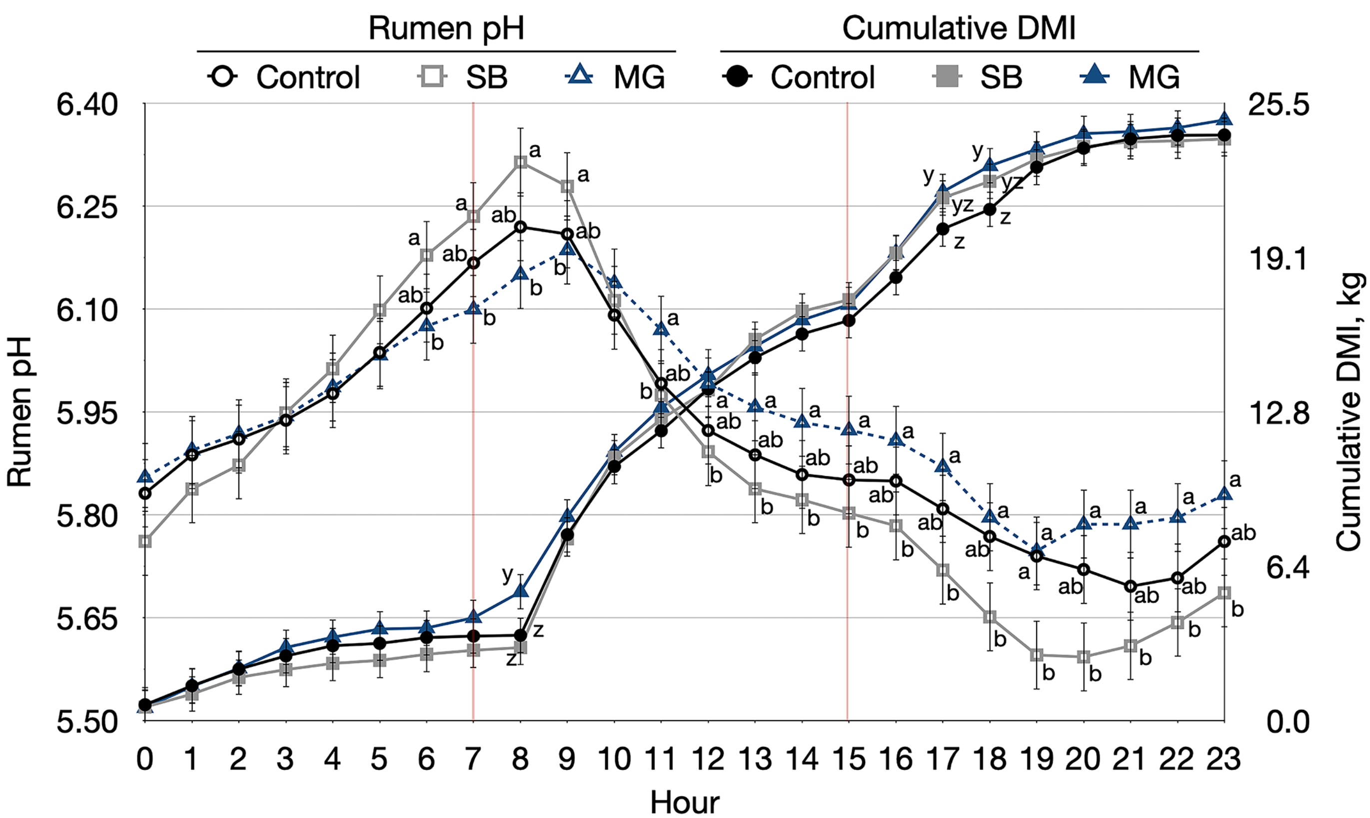

Figure 9. Effect of neutralizing agent and dietary forage-to-concentrate ratio on rumen pH and cumulative DMI throughout the day. Vertical lines denote the times of the day when feed was delivered. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within hour indicate differences in rumen pH between treatments at $P < 0.05$. [y,z]Uncommon letters within hour indicate differences in cumulative feed intake between treatments at $P < 0.05$. Error bars denote SEM.

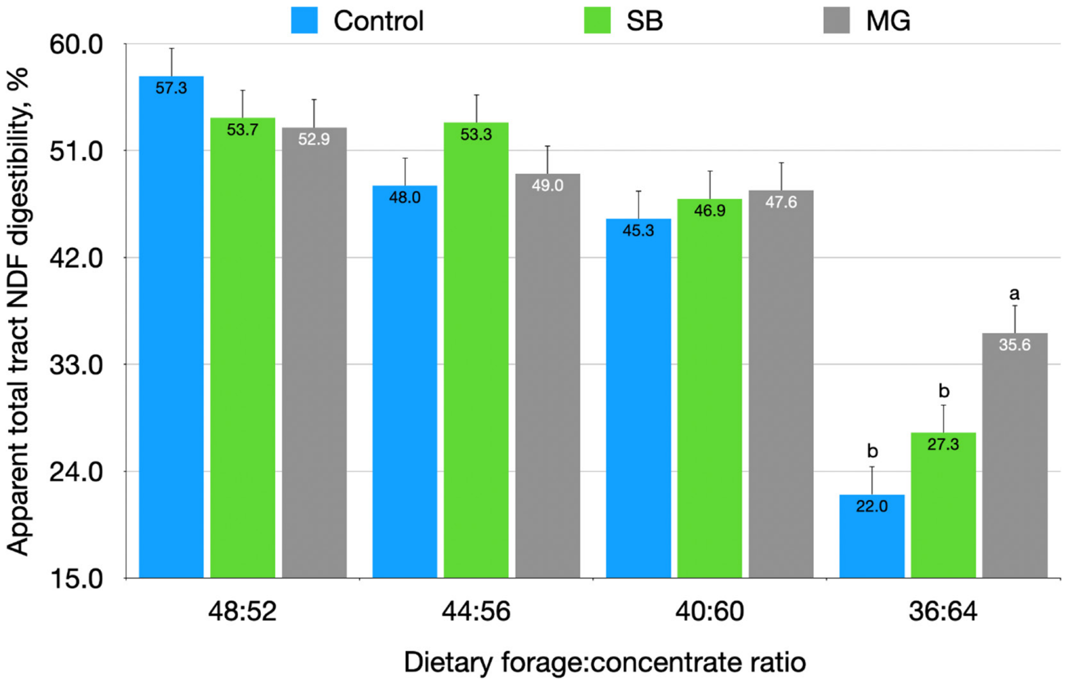

Figure 10. Effect of neutralizing agent and dietary forage-to-concentrate ratio on apparent total-tract NDF digestibility. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within dietary forage-to-concentrate ratio indicate differences between treatments at $P < 0.05$. Error bars denote SEM.

FCR decreased herein is consistent with observations by others (Patra et al., 1993; Stefanska et al., 2018). Although the physiological routes participating in the reduction of plasma Ca consequence of a decrease in rumen pH are not well understood, it has been speculated that Ca may be removed from blood as part of the immune response to facilitate detoxification of endotoxins that could be released due to a low rumen pH (Zebeli et al., 2010). Serum Cl concentrations were greatest in SB and lowest in MG cows, with control cows having intermediate concentrations. Furthermore, serum Cl concentrations increased from 94.9 ± 0.23 to 98.1 ± 0.23 mM as dietary FCR decreased, independently of dietary treatments. This observation disagrees with Li et al. (2012), who reported a tendency for a decreased serum Cl after inducing a rumen acidosis challenge in cattle. Serum P concentrations were unaffected by treatments or periods. Serum Mg concentrations were not affected by the neutralizing agent nor by FCR. However, early studies reported that the addition of sodium bicarbonate depressed plasma Mg concentrations (Erdman et al., 1982; Teh et al., 1985). Serum Na concentrations were lowest in MG cows, and they increased from 136.4 ± 0.24 to 138.2 ± 0.24 mM as dietary FCR decreased, independently of dietary treatment. Increased serum Na concentrations when supplementing cows with sodium bicarbonate has been previously reported (Emery and Brown, 1961), but again, in some studies this increase has not been observed (Escobosa et al., 1984). Also, as observed herein, Patra et al. (1993) reported an increase in serum Na when inducing rumen acidosis. No differences were noted in serum K concentrations among treatments or periods.

In the current study, no differences were observed in serum haptoglobin concentrations (Table 4) across treatments or dietary FCR. Elevated serum haptoglobin concentrations have been linked to inflammation and have been reported to increase with rumen acidosis (Gozho et al., 2005). However, haptoglobin is an acute-phase protein that reacts more slowly and thus reflects the presence of chronic inflammatory conditions (Plaizier et al., 2008). Perhaps the change in rumen pH obtained in this study was not severe enough to elicit a strong inflammation response.

Daily urine output did not differ among treatments (Table 5). Urine Ca concentrations were greatest in control, intermediate in MG, and lowest in SB cows, and they increased from 0.9 ± 0.10 to 1.21 ± 0.10 mM in all cows as dietary FCR decreased, independently of treatment. As a result, urinary Ca excretion was lowest in SB cows, and it increased from 0.96 ± 0.11 to 1.29 ± 0.11 g/d in all cows as dietary FCR decreased, independently of treatment. Again, the increased urinary Ca excretion with decreasing FCR could be linked to an

Table 5. Concentration and excretion of calcium and magnesium in urine as affected by neutralizing agent and dietary forage-to-concentrate ratio

Item	Neutralizing agent[1]			SE	P-value[2]		
	Control	SB	MG		NA	FC	NA×FC
Creatinine, mg/dL	71.0[a]	69.4[a]	59.2[b]	3.22	0.03	0.24	0.18
Urine output, kg/d	30.0	26.8	32.2	2.80	0.52	0.97	0.16
Calcium, m*M*	1.231[a]	0.68[b]	0.9[b]	0.10	<0.01	<0.01	0.31
Calcium output, g/d	1.31[a]	0.69[b]	1.06[a]	0.12	<0.01	<0.01	0.23
Magnesium, m*M*	12.6[b]	12.2[b]	15.8[a]	0.98	<0.01	0.02	0.56
Magnesium output, g/d	7.53[b]	7.40[b]	11.70[a]	0.47	<0.01	0.02	0.11

[a,b]Values within row with uncommon superscripts differ at $P < 0.05$.

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide.

[2]NA = effect of neutralizing agent; FC = effect of dietary forage-to-concentrate ratio; NA×FC = effect of the interaction between dietary neutralizing capacity and forage-to-concentrate ratio.

immune response to decreased rumen pH (Zebeli et al., 2010). However, the lowest urinary Ca excretion in SB cows is consistent with previous studies (Kilmer et al., 1981; Rogers et al., 1985, Tucker et al., 1993) and could be linked to the additional Na supply by sodium bicarbonate (Takagi and Block, 1988). Finally, urine Mg concentrations were greatest in MG cows, which may be linked to the additional Mg provided in this diet. In addition, urine Mg concentration increased from 13.1 ± 0.69 to 14.6 ± 0.69 m*M* as dietary FCR decreased, independently of treatment; consequently, urinary Mg excretion increased from 8.4 ± 0.37 to 9.2 ± 0.37 g/d as dietary FCR decreased. The increased Mg urinary excretion may be a result of an increased absorption of dietary Mg in the rumen because dietary concentration of K diminished as dietary FCR decreased, and elevated rumen K concentrations inhibit the active absorption of Mg (Leonhard-Marek et al., 2010).

Rumen Microbiome

Alpha diversity was unaffected by dietary FCR, but it was reduced ($P < 0.05$) in SB cows independently of the dietary FCR. Alpha diversity, measured as the inverse Simpson index, was greatest ($P < 0.1$) in MG (9.56 ± 0.30) and control (8.95 ± 0.31) cows, whereas SB cows had the lowest diversity (7.84 ± 0.30), independent of dietary FCR (Figure 11). Few studies have evaluated the potential effects of rumen pH modulators on the rumen microbiome. Ramos et al. (2021) reported a numerically progressive increase in α-diversity as the proportion of magnesium oxide in the diet increased, and several studies have described a decrease in α-diversity under subacute rumen acidosis (Fernando et al., 2010; Mao et al., 2013; Petri et al., 2013), although others have reported no change (McGovern et al., 2020). Table 6 depicts β-diversity values (calculated as distance from the group centroid) in the rumen microbiome. Herein, a high β-diversity indicates a low degree of similarity in specific taxa between individuals within a given dietary treatment or dietary FCR. The rumen microbiome at all evaluated taxa levels (from phylum to genus) was more similar (i.e., lower β-diversity) among individuals within the MG treatment than among individuals in control and SB treatments. However, β-diversity increased as dietary FCR decreased, independently of the type of neutralizing agent fed, indicating that as the concentrate challenge increased the dissimilarity in microbial populations among individuals within dietary treatment (in all evaluated taxa) increased, which could be due to different host responses to the dietary challenge.

Both the neutralizing agent and the dietary FCR affected the relative abundance of different phyla in the rumen (Figure 12). The relative abundance of *Bacteroidetes* was greatest in the rumen of control and SB cows, and fluctuated across dietary FCR, with lowest values observed at 44:45 (Figure 12). High rumen relative abundances of *Bacteroidetes* have been linked with high FE in dairy cattle (Delgado et al., 2019). Rumen relative abundance of *Firmicutes* was not affected by the type of neutralizing agent, but it fluctuated across dietary FCR (Figure 12). Rumen relative abundance of *Fibrobacteres*, a phylum that contains a number of fibrolytic bacteria, was greatest in MG cows, and their abundance progressively decreased as dietary FCR decreased, independently of the neutralizing agent (Figure 12). Bacteria within this phylum have been reported to be sensitive to low pH (Li et al., 2017). Within the *Fibrobacteres* phylum, the genus *Fibrobacter* was more (adjusted $P < 0.05$) abundant in MG than in SB cows regardless of dietary FCR. Also, within the *Spirochaetota* phylum, *Treponema* was more (adjusted $P < 0.05$) abundant in MG than in SB cows regardless of dietary FCR. These 2 genera are heavily involved in fiber degradation in the rumen, and their

Table 6. Effects of neutralizing agent and dietary forage-to-concentrate ratio (FCR) on β-diversity (average distance to centroid) of the rumen microbiome, calculated as distance from group centroids

	Neutralizing agent[1]			Dietary FCR				*P*-value[2]		
Taxon	Control	SB	MG	48:52	44:56	40:60	36:64	NA	FC	NA×FC
Phylum	6.14[a]	6.24[a]	5.65[b]	5.37[y]	5.90[xy]	6.21[x]	6.50[x]	0.01	0.02	0.83
Class	8.34[a]	8.44[a]	7.77[b]	7.31[y]	8.19[x]	8.49[x]	8.60[x]	<0.01	<0.01	0.79
Order	12.93[a]	12.64[ab]	12.09[b]	11.72[y]	12.51[xy]	12.75[x]	13.0[x]	<0.01	<0.01	0.36
Family	17.65[a]	17.19[ab]	16.61[b]	16.28[y]	17.09[xy]	17.47[xy]	17.60[x]	<0.01	<0.01	0.41
Genus	23.65[a]	22.69[ab]	21.99[b]	22.02[y]	22.64[xy]	23.14[xy]	23.50[x]	<0.01	<0.01	0.50

[a,b]Values within rows under neutralizing capacity with uncommon superscripts differ at $P < 0.05$.

[x,y]Values within rows under dietary FCR with uncommon superscripts differ at $P < 0.05$.

[1]Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide.

[2]*P*-values correspond to a permanova analysis. NA = effect of neutralizing agent; FC = effect of dietary FCR; NA×FC = effect of the interaction between dietary neutralizing capacity and FCR.

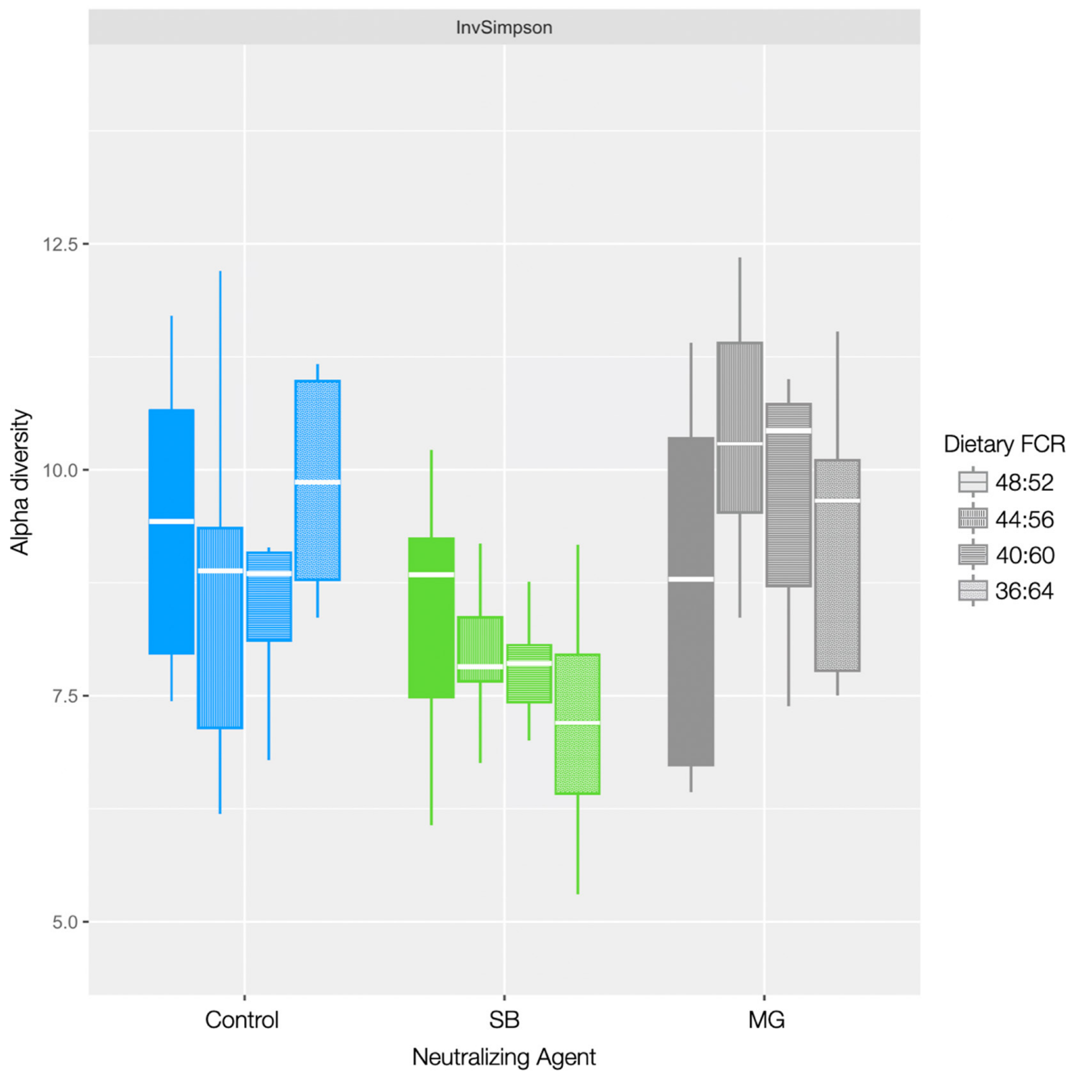

Figure 11. Alpha diversity (measured as inverse Simpson index, InvSimpson) as affected by neutralizing agent and dietary forage-to-concentrate ratio (FCR). Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. The lower and upper edges of each box plot depict the first and third quartiles, the midline shows the median, and the whiskers extend from the minimum to the maximum values.

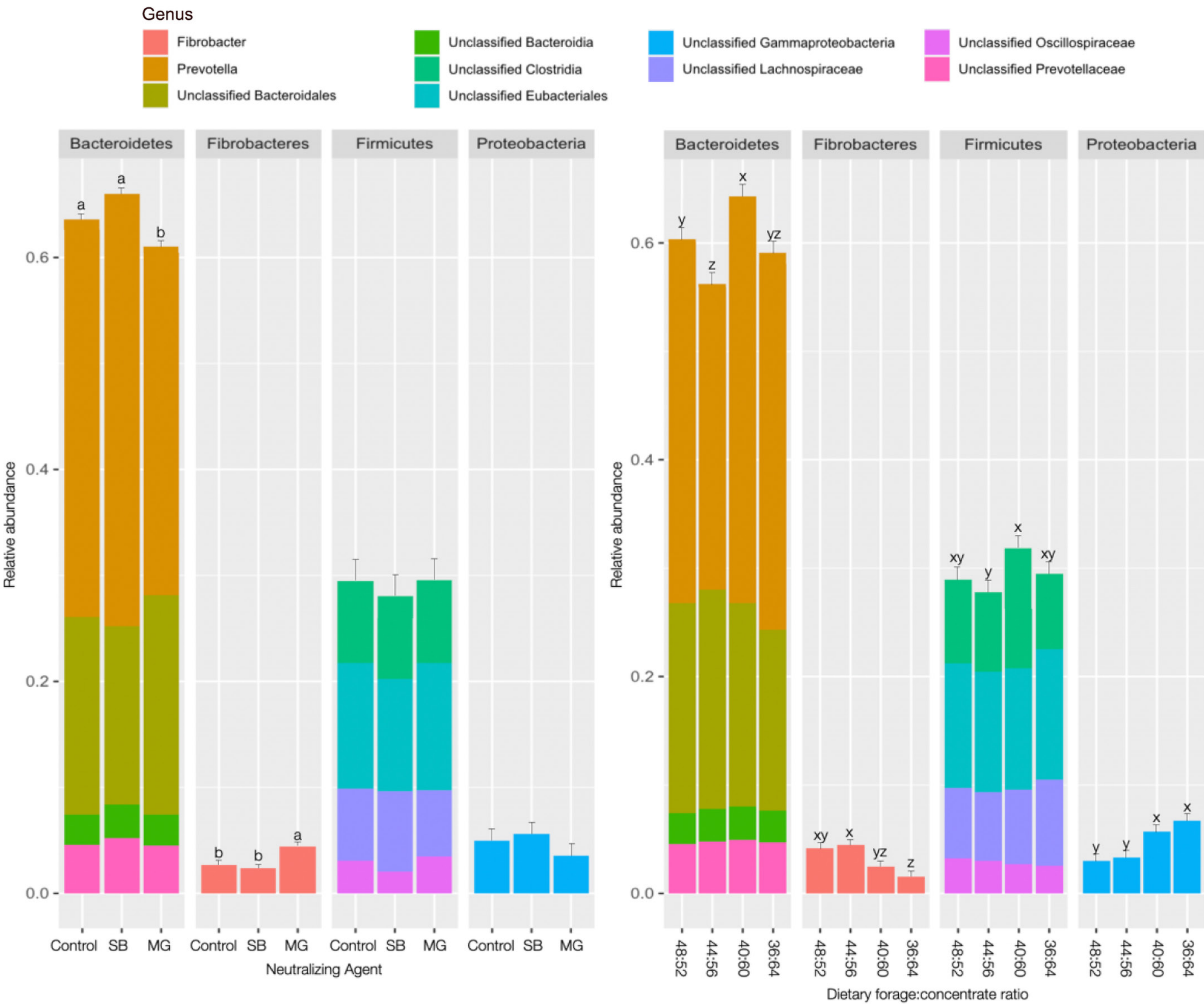

Figure 12. Relative abundance of *Bacteroidetes*, *Fibrobacteres*, *Firmicutes*, and *Proteobacteria* as affected by neutralizing agent and dietary forage-to-concentrate ratio. Control = no supplementation; SB = supplementation of 0.82% (DM basis) of sodium bicarbonate; MG = supplementation of 0.25% of a blend of magnesium oxide. [a,b]Uncommon letters within phylum indicate differences among neutralizing agents at $P < 0.05$. [x–z]Uncommon letters within phylum indicate differences among dietary forage-to-concentrate ratios at $P < 0.05$. Error bars denote SEM.

abundances are typically elevated in high-forage diets (Petri et al., 2013). The greater abundance of *Fibrobacter* and *Treponema* in MG than in SB cows could be one of the reasons why MG cows had greater total-tract NDF digestibility than control and SB cows as dietary FCR decreased (Figure 10). Finally, relative rumen abundance of *Proteobacteria* increased with dietary FCR of 40:60 and 36:64, regardless of the neutralizing agent in the diet (Figure 12). Similar observations have been reported when comparing low- versus high-grain diets. For example, Petri et al. (2013) reported a 20% increase in the abundance of *Proteobacteria* 12 h after a rumen acidosis challenge.

Regarding amylolytic bacteria, they were not affected by the neutralizing agent, but their relative abundances were, in general, increased as dietary FCR decreased. The genus *Weissella* (a lactic-acid producing microorganism) progressively increased (adjusted $P < 0.05$) as dietary FCR decreased independently of treatment. Similarly, the relative abundances in the rumen of *Selenomonas* and *Butyrivibrio*, 2 important amylolytic genera, were greater when cows consumed a dietary FCR of 40:60 than an FCR of 44:56 and greater with a dietary FCR of 36:64 than an FCR of 44:56, independently of treatment. Lastly, relative rumen abundances of *Ruminococcus*, another important amylolytic genus, were also greater when feeding a dietary FCR of 36:64 than an FRC of 48:52.

CONCLUSIONS

With progressive reductions in dietary FCR, feed intake tended to decrease and ECM increased, independent of dietary treatments. When providing dietary FCR of 44:56 or 40:60, feed intake was lower in cows receiving no neutralizing agent or supplemented with sodium bicarbonate than in cows supplemented with magnesium oxide. Changes in intake are likely due to rumen conditions, as rumen pH decreased with progressive reduction of dietary FCR. Apparent NDF digestibility decreased as dietary FCR decreased, and at low dietary FCR, apparent total-tract NDF digestibility was lower in cows unsupplemented or supplemented with sodium bicarbonate than in those supplemented with magnesium oxide. Changes in fiber digestibility may be partly due to modifications in the rumen microbial population. Despite the fact that the theoretical NC of a diet is similar when removing sodium bicarbonate and replacing it by a blend of magnesium oxide at one-third of the dose of sodium bicarbonate, rumen pH was increased, especially when dietary FCR was low. When assessing the potential effects of the dietary cation-anion balance on feed intake when feeding magnesium oxide and removing sodium bicarbonate, it is preferable to calculate the dietary anionic balance accounting for Ca, P, and Mg in addition to Na and K.

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